

HyProSim

Hybrid Rocket Engine Propulsion
Simulation, an Engineering Tool

Project HyProSim



Porto Space Team



Porto Space Team
FEUP Universidade do Porto

Preface

This document serves as the comprehensive manual and technical documentation for HyProSim, an advanced engineering tool developed in Python with a Java interface for the simulation of Hybrid Rocket Engines. In an era where efficient and reliable space access is paramount, the complexities inherent in multi-physics propulsion systems necessitate sophisticated analytical and predictive capabilities. HyProSim addresses this critical need by providing a robust platform for the accurate modeling and simulation of hybrid rocket engine performance.

The primary objective of this manual is to furnish users, from students to experienced engineers, with a thorough understanding of HyProSim's functionalities, underlying methodologies, and practical applications. It is meticulously structured to guide the reader through the theoretical foundations of hybrid propulsion, the architectural design of HyProSim, its operational procedures, and methods for interpreting simulation results. Furthermore, this documentation highlights the tool's contribution to reducing the reliance on costly physical testing, thereby accelerating the development cycle of next-generation propulsion technologies.

As the global aerospace sector pivots towards more sustainable and accessible space solutions, tools like HyProSim are indispensable. Their capacity to enhance design optimization, ensure operational safety, and foster innovation is crucial for the advancement of rocket propulsion. This manual is therefore an essential companion for anyone seeking to leverage HyProSim to its full potential in contributing to the future of space exploration.

Collaborators:

Rafael Lino,

Pedro Gameiro,

Leonor Gonçalves,

Ilídio Costa

Porto Space Team, September 2025

Contents

Preface	ii
Nomenclature	vii
1 Introduction	1
1.1 Hybrid Propulsion in Portugal	1
1.2 Similar works	2
1.3 Sustainable Development Goals	2
1.4 HyProSim Mascot	3
1.5 Scope	3
2 Simulator Overview	4
2.1 Hybrid Rocket Engine Modules	5
2.2 Units of Every Sub-System	5
3 Methodology & Fundamentals	7
3.1 Oxidizer Tank	7
3.1.1 Auto-Pressurized Tank	7
3.1.2 Internally Pressurized Tank	8
3.1.3 Externally Pressurized Tank via Pressure Vessel	8
3.2 Injector	9
3.2.1 Pressure Losses	9
3.2.2 Shower-Head Injector	9
3.3 Oxidizer mass flow	10
3.3.1 Incompressible Viscous Fluid Model	10
3.3.2 Homogeneous Equilibrium Model	10
3.3.3 Non Homogeneous Non Equilibrium Model	10
3.3.4 Solomon NHNE Model Correction	11
3.3.5 Gas Choked Mass Flow Model	11
3.4 Fuel mass flow	12
3.4.1 Standard Regression Model	12
3.4.2 Hills Lattice-Augmented Regression Model	12
3.5 Geometric properties of the grain	21
3.5.1 Mathematical model	21
3.5.2 Practical application	23
3.6 Fuel Combustion Mass Flow	27
3.6.1 Fuel Combustion Mass Flow Model	27
3.6.2 Chemical Combustion Mass Flow	27
3.6.3 Fuel Mass Fraction Update	28
3.6.4 Ignition Process and O ₂ Release	29
3.6.5 Available Surface Heating Power	29
3.6.6 Effective Vapor Mass Flow Entering the Chamber	30
3.6.7 Fuel Combustion Mass Flow Model Fluxogram	30
3.7 Combustion Chamber	31
3.7.1 Mass Transient Mathematical Model	32
3.7.2 Volume Transient Mathematical Model	32
3.7.3 Pressure Transient Mathematical Model	33
3.7.4 Temperature Transient Mathematical Model	33
3.8 Nozzle	36
3.8.1 Nozzle Mass Flow	36

4	Implementation	37
4.1	Grain Geometry as applied	37
4.1.1	Variables	38
5	Results & Analysis	41
5.1	INVICTUS 1.0 - Hot Fire Test 27/09/2024	41
5.1.1	Motor Detail	41
5.1.2	Experimental results	42
5.1.3	Simulated results	43
6	Conclusion	45
	References	46

List of Figures

2.1	Hybrid Rocket Motor Modules overview	5
3.1	Auto-pressurized tank illustration	7
3.2	Externally Pressurized Tank and Vessel Pressure over time with a constant mass flow .	8
3.3	Injector illustration with consequent parameters.	11
3.4		12
3.5		14
3.6		14
3.7		15
5.1	Invictus 1.0 hybrid rocket motor testing setup and Rocket Piping and Instrumentation Diagram.	41
5.2	Combustion Chamber of INVICTUS 1.0	42
5.3	Hot Fire Test (27/09/2024) data; a) Top & bottom tank pressure [bar]; b) Combustion chamber pressure [bar]; c) Thrust [N].	43
5.4	Full 36s thrust curve	43
5.5	Comparison between experimental (exp.) and simulated (sim.) results from INVICTUS 1.0 hot fire test.	44

List of Tables

2.1	SI Base Units	6
2.2	SI Derived Units with Special Names and Symbols	6
3.1	Regression law coefficient and flux exponent for different fuel and oxidizer combinations	12
5.1	Nozzle geometry	42

Nomenclature

If a nomenclature is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Abbreviations

Abbreviation	Definition
ISA	International Standard Atmosphere
...	

Symbols

Symbol	Definition	Unit
T_c	Chamber Temperature	[kg/s]
P_c	Chamber Pressure	[kg/s]
P_t	Tank Pressure	[kg/s]
P_v	N_2O Saturation Pressure	[kg/s]
$P_{inj.}$	Injector Pressure	[kg/s]
h_{in}	Reactants Enthalpy	[J/kg]
h_{out}	Products Enthalpy	[J/kg]
$\Delta h_{comb.}$	Combustion Enthalpy	[J/kg]
h_{in}	Reactants Enthalpy	[J/kg]
$\dot{m}_{ox}$	Oxidizer Mass Flow	[kg/s]
$\dot{m}_{fuel}$	Grain Mass Regression Flow	[kg/s]
$\dot{m}_{comb,stock}$	Oxidizer Mass Flow	[kg/s]
$\dot{m}_{comb,inlet}$	Oxidizer Mass Flow	[kg/s]
$\dot{m}_{comb,fuel}$	Oxidizer Mass Flow	[kg/s]
$\dot{m}_{comb.mix}$	Oxidizer Mass Flow	[kg/s]
$\dot{m}_{vap,in}$	Oxidizer Mass Flow	[kg/s]
$f_{vap.}$	Oxidizer Mass Flow	[kg/s]

Symbol	Definition	Unit
f_{burn}	Oxidizer Mass Flow	[kg/s]
f_{ent}	Oxidizer Mass Flow	[kg/s]
$\eta_{comb.}$	Oxidizer Mass Flow	[kg/s]

1

Introduction

HyProSim arose from project INVICTUS, which encompasses the development of a high-performance hybrid-propulsion rocket with a apogee of 3 km aimed at participating in the European Rocketry Challenge (EuRoC). Due to the high cost of experimental testing, simulation of these complex multi-physics hybrid rocket engines proves to be essential. In 2024, Porto Space Team marked history as the first Portuguese student-led team to produce and validate a hybrid rocket engine and first to compete in Euroc with a Portuguese hybrid engine.

Hybrid rocket engines are of significant relevance to the future of space exploration, particularly for Portugal. This technology offers a compelling combination of safety and controllability, making it ideal for launching small satellites, a rapidly expanding market segment. Furthermore, hybrid engines present a more sustainable alternative, aligning with Portugal's environmental objectives.



1.1. Hybrid Propulsion in Portugal

The Portuguese Air Force's program NORTH, a collaborative effort involving the Portuguese Air Force, the University of Minho, and Omnidea, underscores Portugal's commitment to advancing this technology. Additionally, the upgrades to the Santa Maria station will provide critical support for these launches. These initiatives are integral to Portugal's broader space strategy, as exemplified by New Space Portugal, which seeks to establish the nation as a prominent player in the global space sector.

The relevance of hybrid propulsion has increased significantly in recent years due to its combination of safety, controllability, and environmental benefits. This has led to its adoption in various projects, such as the development of micro-launchers by companies like HyImpulse, which aim to provide cost-effective and flexible launch solutions for small satellites. The growing demand for sustainable and efficient space transportation has further fuelled interest in hybrid propulsion systems globally.

The ability to accurately understand and simulate hybrid rocket engines is crucial for advancing space exploration. This knowledge will enable us to optimize engine performance, improve safety, and develop more efficient and sustainable launch systems. As we continue to explore the vast potential of space, these advancements will undoubtedly play a pivotal role in shaping the future of space travel.

1.2. Similar works

There are works which delve in the simulation of hybrid rocket engines, due to previously stated advantages. This can be resumed to the following list of project:

- **MDFRSV** (Multidisciplinary Design Framework for Reusable Space Vehicles) is being developed by Instituto Superior Técnico (IST) within the New Space Portugal project scope, is composed of 5 key building blocks: propulsion, mass & sizing, trajectory, aerodynamics, and life cycle assessment <https://doi.org/10.3390/aerospace11020126>.
- **Project Chimera**, developed by Delft Aerospace Rocket Engineering (DARE), aims to develop new and innovative hybrid rocket engines <https://dare.tudelft.nl/projects/>.
- **AHRES** (Advanced Hybrid Rocket Engine Simulation), developed by DLR - Institute of Aerodynamics and Flow Technology, enables the optimised design of a complete hybrid or solid rocket engine <https://www.dlr.de/en/as/research-and-transfer/software-solutions/aerothermodynamics/software-ahres>.
- **HRAP** (The Hybrid Rocket Analysis Program) was developed by Robert Nickel for use by the University of Tennessee Rocket Engineering Team, readily mentioned in only amateur rocketry forums. https://github.com/rnickel1/HRAP_Source.
- **HRPC** (The Hybrid Rocket Performance Code), developed by University of KwaZulu-Natal, is a predictive motor performance code based on a one-dimensional unsteady gas flow model through the nozzle and, on a zero-dimensional model of the combustion chamber capturing only the filling and emptying dynamics https://www.scielo.org.za/scielo.php?script=sci_arttext&pid=S2309-89882017000100005

1.3. Sustainable Development Goals

The Sustainable Development Goals (SDGs) are a global blueprint for a sustainable future, addressing issues from poverty to climate change and fostering international cooperation to achieve a better world by 2030. The HyProSim project aligns with key SDGs through its innovative engineering tool, namely it directly supports:

- **SDG 4 - Quality Education:** The project is developed by a student-led team and its documentation is designed to be a comprehensive manual for students and experienced engineers alike, serving as a tool to enhance understanding of hybrid propulsion systems. This educational aspect is critical for fostering the next generation of engineers and scientists who will continue to advance rocket propulsion and space exploration.
- **SDG 9 - Industry, Innovation and Infrastructure:** It promotes innovation and the development of new propulsion technologies for the aerospace industry. By reducing reliance on expensive physical tests, HyProSim accelerates the development cycle of next-generation propulsion systems, which are crucial for efficient and reliable space access.
- **SDG 13 - Climate Action:** Hybrid Propulsion technology is a more sustainable alternative for space solutions, aligning with Portugal's environmental objectives and the global pivot toward more accessible and sustainable space solutions.



1.4. HyProSim Mascot

The Iberian Desman ("*Toupeira-da-água*" in Portuguese), our project's mascot, is a unique and endangered semi-aquatic mammal, perfectly reflecting HyProSim's hybrid nature. Just as the Desman thrives in both land and water, our tool simulates the complex interplay of solid and liquid states in hybrid rocket engine propulsion. By featuring this special creature, we aim to raise awareness about its conservation needs and highlight the importance of preserving biodiversity. We hope our work contributes to the ongoing efforts to protect this fascinating species.

The Iberian Desman is a small, mole-like mammal that lives in rivers and streams in Portugal and Spain. It's one of the few venomous mammals, and its saliva contains a neurotoxin that can paralyse its prey. The Iberian Desman is also an endangered species, and its population has declined significantly in recent years due to habitat loss and pollution. Conservation efforts are underway to protect the Iberian Desman and its habitat, such as ICNF and IUCN.

1.5. Scope

HyProSim aims to be an engineering tool to predict and optimize hybrid rocket engines, developed by students, fostering their understanding of hybrid chemical propulsion and programming.